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Bode et al.

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(54) **MULTI-COLOR HOT STAMP PRINTING SYSTEM**

(71) Applicant: **Illinois Tool Works Inc.**, Glenview, IL (US)

(72) Inventors: **Kenneth Bode**, Philadelphia, PA (US); **Michael Colella**, Southington, CT (US); **Daniela Israel**, Glenview, IL (US)

(73) Assignee: **ILLINOIS TOOL WORKS INC.**, Glenview, IL (US)

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(60) Provisional application No. 63/086,869, filed on Oct. 2, 2020, provisional application No. 63/040,834, filed on Jun. 18, 2020.

(51) **Int. Cl.**
B41K 3/24 (2006.01)
B41K 3/58 (2006.01)

(52) **U.S. Cl.**
CPC . ***B41K 3/58*** (2013.01); ***B41K 3/24*** (2013.01)

(58) **Field of Classification Search**
CPC B41F 16/0046; B41K 3/24; B41K 3/58; B41M 1/14
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,961,121	A	6/1976	Warsager
4,250,511	A	2/1981	Stein et al.
5,683,785	A	11/1997	Talvalkar et al.
5,985,078	A	11/1999	Suess et al.
6,152,621	A	11/2000	Langan
9,290,038	B2	3/2016	Colella

(Continued)

FOREIGN PATENT DOCUMENTS

CN	1963571	A	5/2007
CN	102214412	A	10/2011

(Continued)

OTHER PUBLICATIONS

International Preliminary Report on Patentability mailed Feb. 22, 2022 for corresponding PCT/US2021037445 (8 pages).

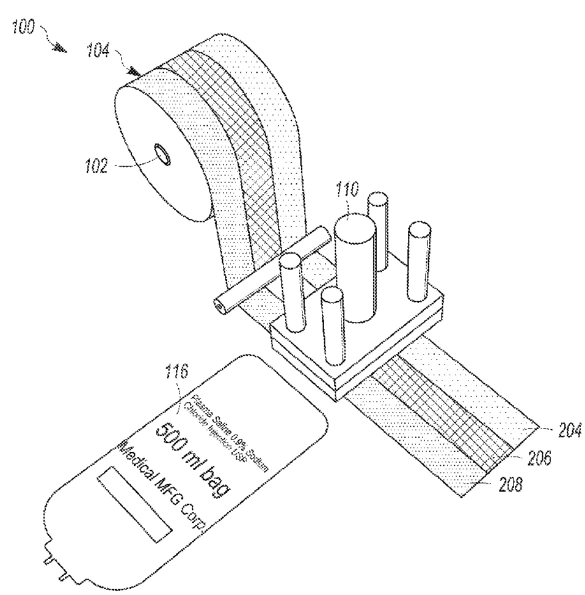
(Continued)

Primary Examiner — Christopher E Mahoney
Assistant Examiner — Marissa Ferguson-Samreth
(74) *Attorney, Agent, or Firm* — The Small Patent Law Group LLC; Christopher R. Carroll

(57) **ABSTRACT**

A method includes moving a multi-color ribbon over a target object. The multi-color ribbon has a lower layer and an upper layer of inks or pigments. The method also includes applying heat and pressure onto the multi-color ribbon to transfer the overlying layers of the inks or pigments to the target object, and removing at least a portion of the lower layer of the ink or pigment that was transferred to the target object to reveal at least a portion of the upper layer of the ink or pigment that were transferred to the target object.

22 Claims, 5 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

11,065,910	B2	7/2021	Wei et al.	
11,279,144	B2 *	3/2022	Sangkaratana	B41M 7/0054
2009/0074948	A1 *	3/2009	Ihara	B41M 5/0256 427/8
2019/0193388	A1	6/2019	Fisher et al.	
2019/0378438	A1	12/2019	Penrose et al.	
2021/0179894	A1	6/2021	Wei et al.	

FOREIGN PATENT DOCUMENTS

EP	0537668	A2		4/1993	
GB	2199538	A		7/1988	
GB	2270392	A		3/1994	
JP	2003-237247		*	8/2003	
WO	1999025562	A1		5/1999	
WO	WO-0068015	A2	*	11/2000	B41J 15/16
WO	WO-2020234386	A1	*	11/2020	A61J 1/10

OTHER PUBLICATIONS

International Search Report and Written Opinion mailed Sep. 20, 2021 for corresponding PCT/US2021037445 (11 pages).

International Search Report and Written Opinion mailed Feb. 22, 2022 for corresponding application PCT/US21/53068 (12 pages).

International Preliminary Report on Patentability mailed Apr. 13, 2023 for corresponding PCT/US2021/053068 (8 pages).

* cited by examiner

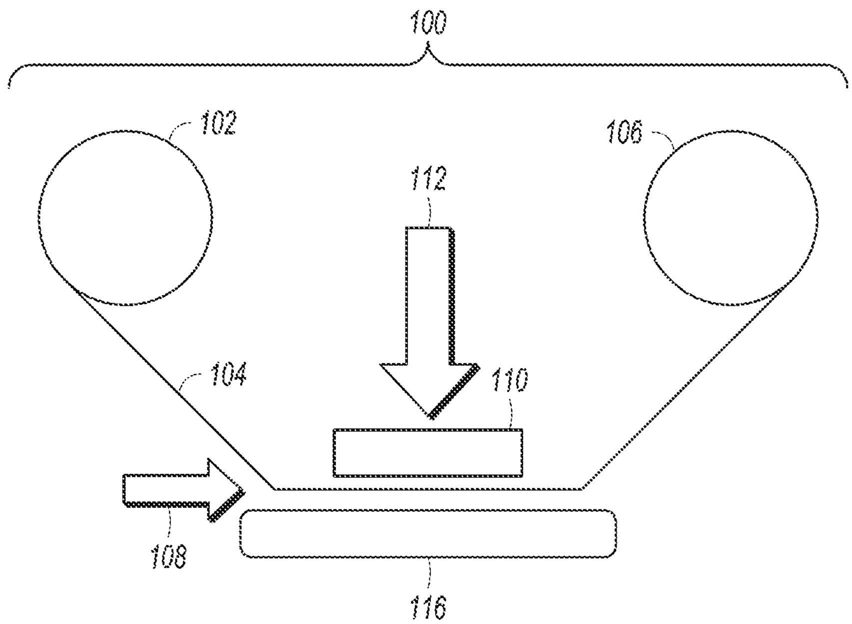


FIG. 1

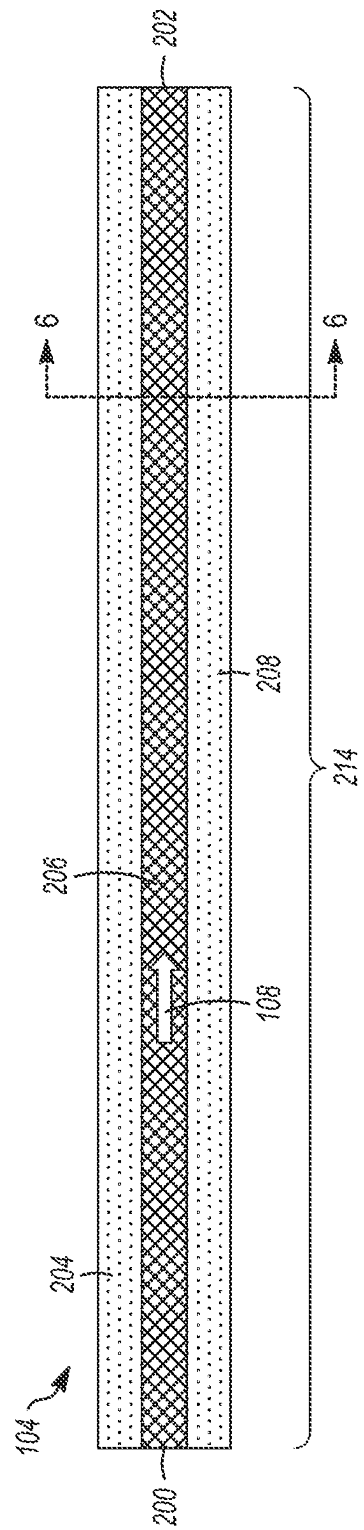

202
6
6
104
204
206
108
208
214
200

FIG. 2

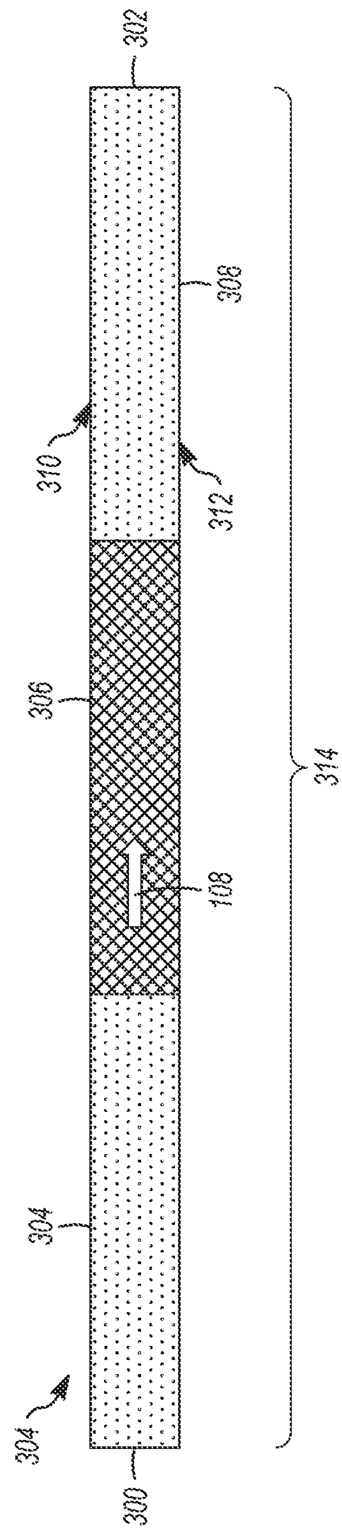

302
308
310
312
306
108
314
304
304
300

FIG. 3

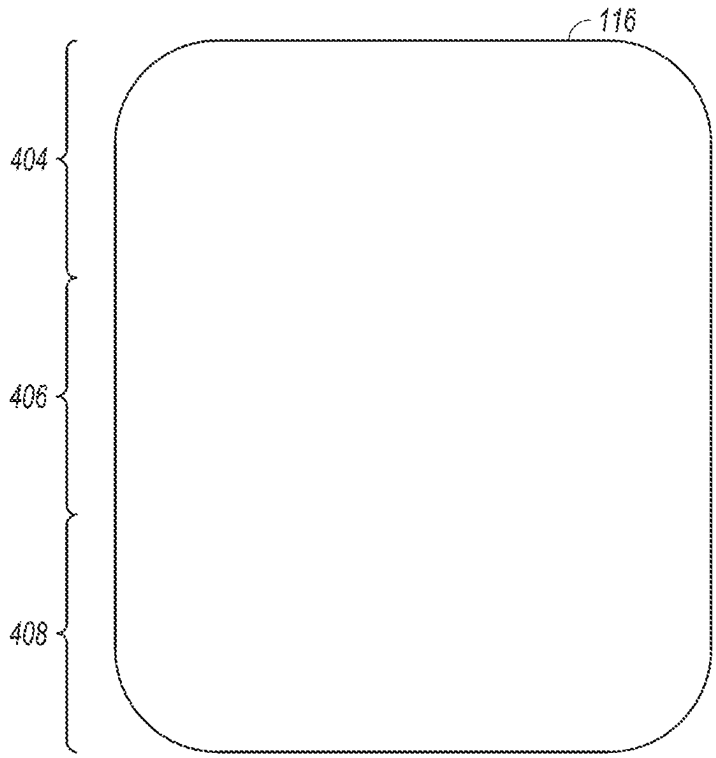

116
404
406
408

FIG. 4

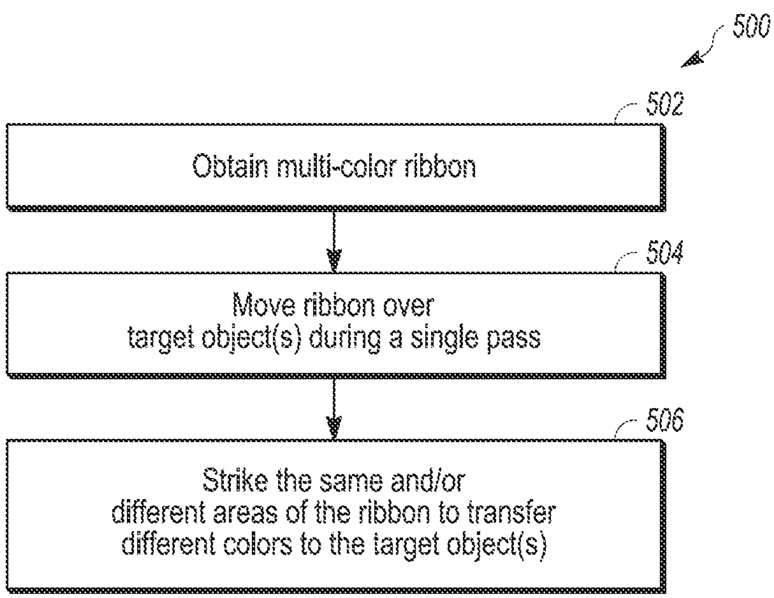

500
502
Obtain multi-color ribbon
504
Move ribbon over
target object(s) during a single pass
506
Strike the same and/or
different areas of the ribbon to transfer
different colors to the target object(s)

FIG. 5


600
214
602

FIG. 6

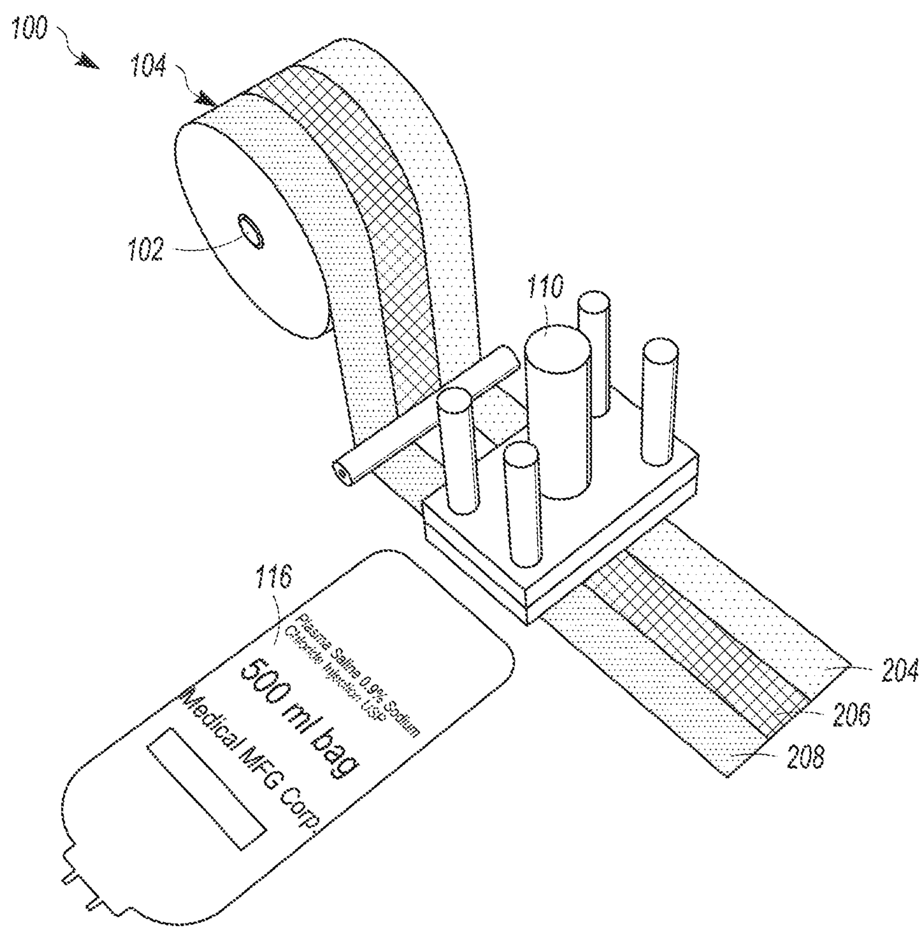

100
104
102
110
116
Plasma Saline 0.9% Sodium
Chloride Injection USP
500 ml bag
Medical MFG Corp.
204
206
208

FIG. 7

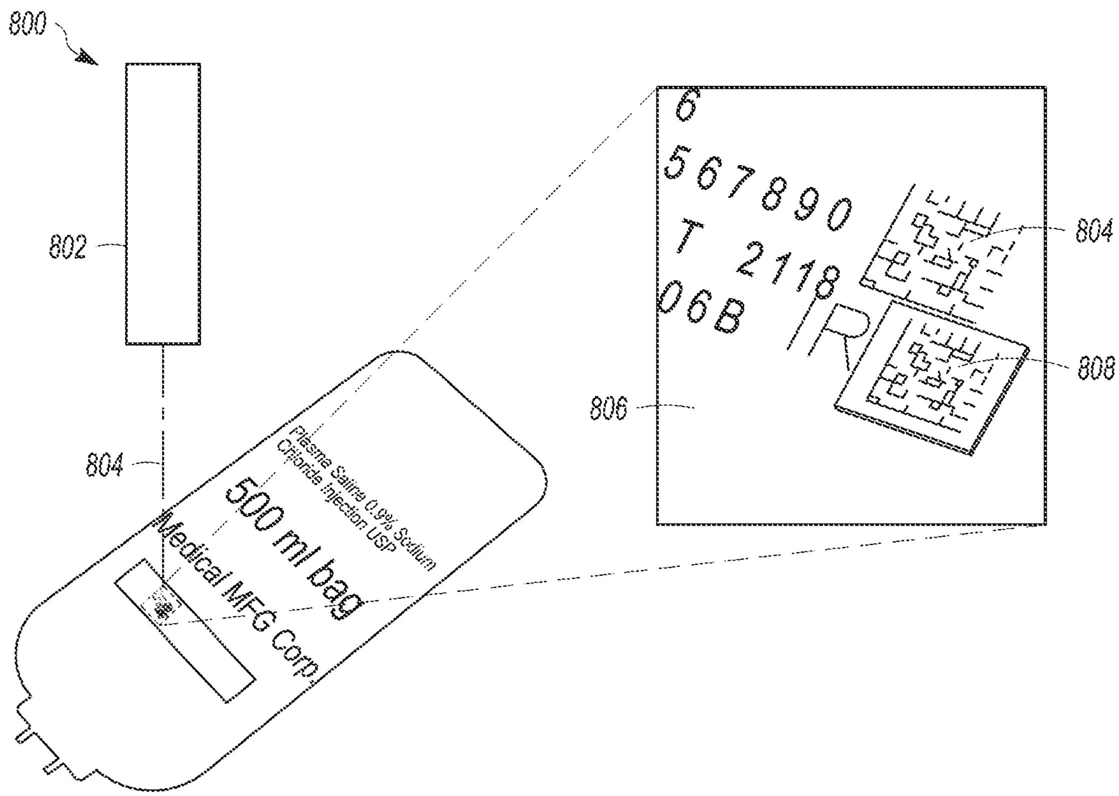

800
802
804
Plasma Saline 0.9% Sodium
Chloride Injection USP
500 ml bag
Medical MFG Corp
6
567890
T 2118
06B
804
808
806

FIG. 8

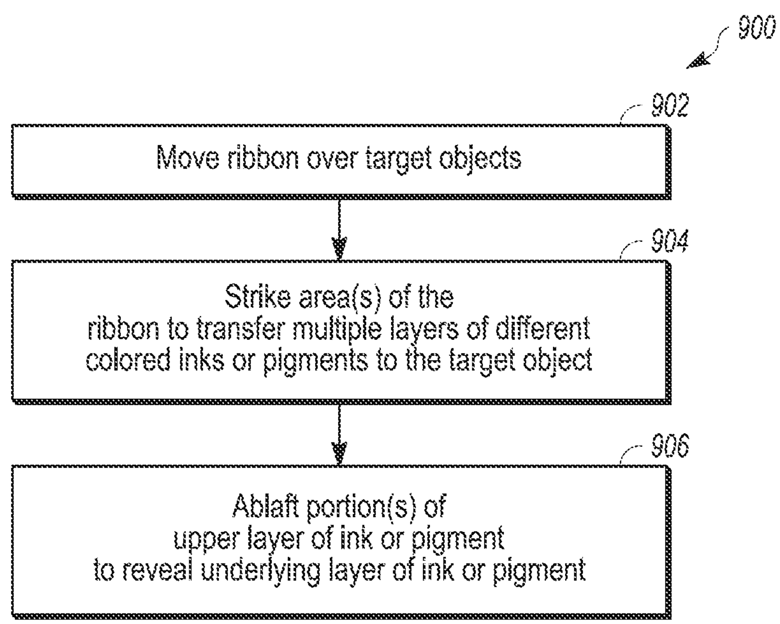

900
902
Move ribbon over target objects
904
Strike area(s) of the
ribbon to transfer multiple layers of different
colored inks or pigments to the target object
906
Ablaft portion(s) of
upper layer of ink or pigment
to reveal underlying layer of ink or pigment

FIG. 9

MULTI-COLOR HOT STAMP PRINTING SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Application No. 63/086,869 (filed 2 Oct. 2020), and is a continuation-in-part of International Patent Application No. PCT/US2021/037445 (filed 15 Jun. 2021), which claims priority to U.S. Patent Application No. 63/040,834 (filed 18 Jun. 2020). The entire disclosures of these applications are incorporated herein by reference.

BACKGROUND

Technical Field

The subject matter described herein relates to hot stamp printing systems.

Discussion of Art

Some known printing ribbons used in hot stamping systems provide a single color that is transferred onto an article to be printed on by a die. One problem with this technique is that only a single color may be used to print on the article. To stamp multiple different colors on the article, however, multiple different ribbons having the different colors usually must be used. This can significantly lengthen the time needed to decorate on the target object as multiple rolls of the ribbons may need to be used and same article may need to be printed upon multiple times or with multiple stampings.

BRIEF DESCRIPTION

In one embodiment, a method includes moving a multi-color ribbon over a target object. The multi-color ribbon has a lower layer and an upper layer of inks or pigments. The method also includes applying heat and pressure onto the multi-color ribbon to transfer the overlying layers of the inks or pigments to the target object, and removing at least a portion of the lower layer of the ink or pigment that was transferred to the target object to reveal at least a portion of the upper layer of the ink or pigment that were transferred to the target object.

In one embodiment, a multi-color ribbon configured for use in a hot stamp printing system is provided. The ribbon includes an elongated carrier that continuously extends from a first end to an opposite end, and multiple, differently colored inks or pigments disposed in different layers of the elongated carrier between the first end and the second end of the elongated carrier. The elongated carrier is configured to be moved across a target object with heat and pressure applied to the elongated carrier to transfer two or more of the differently layers of the differently colored inks or pigments to the target object for hot stamp printing of the two or more layers of the differently colored inks to the target object.

In one embodiment, a hot stamp printing system includes a heated die configured to strike a multi-colored ribbon having overlying layers of different color inks or pigments to transfer the layers of the different color inks or pigments to a target object, and an ablation assembly configured to generate an ablation energy toward the layers of the different color inks or pigments on the target object, remove at least a portion of an upper layer of the layers of the different color inks or pigments on the target object, and reveal an underlying layer of the layers of the different color inks or pigments on the target object.

BRIEF DESCRIPTION OF THE DRAWINGS

The inventive subject matter may be understood from reading the following description of non-limiting embodiments, with reference to the attached drawings, wherein below:

FIG. **1** schematically illustrates a hot stamp printing system;

FIG. **2** illustrates one example of a hot stamp multi-color ribbon shown in FIG. **1**;

FIG. **3** illustrates another example of a hot stamp multi-color ribbon;

FIG. **4** schematically illustrates a top view of one example of a target object shown in FIGS. **1** and **2**;

FIG. **5** illustrates a flowchart of one embodiment of a method for multi-color hot stamp printing;

FIG. **6** illustrates a cross-sectional view of another example of a multi-color ribbon along line **6-6** shown in FIG. **2**;

FIG. **7** illustrates another example of the hot stamp printing system shown in FIG. **1**;

FIG. **8** illustrates one example of an ablation assembly; and

FIG. **9** illustrates a flowchart of one example of a method for hot stamp printing.

DETAILED DESCRIPTION

FIG. **1** schematically illustrates a hot stamp printing system **100**. The printing system **100** includes an unwind reel **102** holding a multi-color ribbon **104**. The ribbon **104** extends from the unwind reel **102** to a wind-up reel **106**. The ribbon **104** is unwound from the unwind reel **102** to the wind-up reel **106** along a web or ribbon direction **108**. The web or ribbon direction **108** represents the direction in which the ribbon **104** moves during printing (e.g., from the unwind reel **102** to the wind-up reel **106**). As the ribbon **104** moves along the web or ribbon direction **108**, a heated die **110** strikes the ribbon **104** along a transfer direction **112**. The pressure and heat of the die **110** transfers at least some of the coatings from the ribbon **104** to a target object **116** to be printed upon, such as an intravenous (IV) bag, a card, a document, or any of a variety of other objects. This stamping onto the target object **116** transfers the pigments or inks that are in the ribbon **104** onto the target object **116**. The inks or pigments may be coatings formed from one or more polymeric binders and one or more colored pigments.

The target object **116** may be one of several target objects **116** that are successively printed on by the printing system **100**. For example, several target objects **116** may move in a machine direction to successively print on the target objects **116**. The machine direction in which the target objects **116** move may be the same as the web or film direction **108** or may be different from the web or film direction **108**. The machine direction may be orthogonal to the web or film direction **108** (e.g., into or out of the plane of FIG. **1**) or at another angle to the web or film direction **108**.

In contrast to some known hot stamp ribbons that have only a single color of ink (coatings) or pigments in the ribbon, the ribbon **104** includes inks or pigments of multiple colors in different areas of the ribbon **104**. Two or more of these colors can be transferred from the ribbon **104** onto the target object **116** during a single strike of the die **110**.

FIG. **2** illustrates one example of the multi-color ribbon **104** shown in FIG. **1**. As shown, the ribbon **104** is an elongated carrier **214** continuously extending from a trailing end **200** to a leading end **202**. The carrier **214** also can be referred to as a film **214** or carrier film **214**. The leading end **202** may be the end of the ribbon **104** that is initially coupled with the wind-up reel **106**. The ribbon **104** may be pulled in the ribbon direction **108** during the printing process described above in connection with FIG. **1**.

The ribbon **104** includes different areas **204**, **206**, **208** that hold different colored inks or pigments. In the illustrated example, the ribbon **104** includes the elongated areas **204**, **206**, **208** that each extend from one end **200** to the opposite end **202**. These areas **204**, **206**, **208** are elongated in the machine direction **108** during hot stamp printing. While three areas **204**, **206**, **208** are shown, alternatively, the ribbon **104** may include two areas (such that the ribbon **104** holds two different colored inks or pigments) or more than three areas (such that the ribbon **104** holds three or more different colored inks or pigments).

FIG. **3** illustrates another example of a multi-color ribbon **304**. The ribbon **304** may be used in place of the ribbon **104** shown in FIGS. **1** and **2** in a hot stamp printing process. The ribbon **304** is an elongated carrier **314** continuously extending from a trailing end **300** to a leading end **302**. The carrier **314** optionally can be referred to as a film **314** or carrier film **314**. Similar to the ribbon **104**, the leading end **302** may be the end of the ribbon **304** that is initially coupled with the wind-up reel **106**. The ribbon **304** may be pulled in the machine direction **108** during the printing process described above in connection with FIG. **1**.

The ribbon **304** includes different areas **304**, **306**, **308** that hold different colored inks or pigments. These areas **304**, **306**, **308** optionally can be referred to as panels. In the illustrated example, the ribbon **304** includes the elongated areas **304**, **306**, **308** that do not extend from one end **300** to the opposite end **302**. For example, the areas **304**, **306**, **308** extend across the entire width of the ribbon **304** from one edge **310** of the ribbon **304** to the opposite edge **312** of the ribbon **304**, with each edge **310**, **312** extending from one end **300** to the other end **302**. Conversely, none of the areas **204**, **206**, **208** in the ribbon **104** extends across the entire width of the ribbon **104** from one edge **310** to the opposite edge **312** as in the ribbon **304**. Instead, the area **204** in the ribbon **104** extends from one edge partially toward (e.g., a third of the way) the opposite edge, the area **206** extends from the area **204** partially toward (e.g., another third of the way) to the opposite edge, and the area **208** extends from the area **206** to the opposite edge. While three areas **304**, **306**, **308** are shown, alternatively, the ribbon **304** may include two areas (such that the ribbon **304** holds two different colored inks or pigments) or more than three areas (such that the ribbon **304** holds three or more different colored inks or pigments).

FIG. **4** schematically illustrates a top view of one example of the target object **116** shown in FIGS. **1** and **2**. The visible side of the target object **116** in FIG. **4** can be the surface that is contacted by the ribbon **104** or **304** when the die **110** strikes the ribbon **104** or **304** to transfer inks or pigments from the ribbon **104** or **304**.

The arrangements of the different inks in the ribbons **104**, **304** can allow or prevent different colors of inks or pigments from being transferred to different areas of the target object **116**. During printing with either ribbon **104**, **304**, a single strike of the die **110** can transfer multiple, different colored inks from the ribbon **104**, **304** to the target object **116**. For example, the die **110** can strike multiple areas **204**, **206**, and/or **208** of the ribbon **104** at the same time to simultaneously transfer different colored inks from the areas **204**, **206**, and/or **208** that are stricken. Because different areas **204**, **206**, **208** of the ribbon **104** move over different areas **404**, **406**, **408** of the target object **116**, however, the ribbon **104** may only be able to transfer certain colors to certain areas **404**, **406**, **408** of the target object **116**.

For example, the ribbon **104** may be able to transfer the color of the ink in the area **204** to a first area **404** of the target object **116** as the area **204** of the ribbon **104** moves over this first area **404** of the target object **116**. The ribbon **104** may not transfer the ink or pigment having another color in the area **206** or **208** to the first area **404** of the target object **116** as neither the area **206** nor the area **208** of the ribbon **104** move over the first area **404** of the target object **116**.

Similarly, the ribbon **104** may be able to transfer the color of the ink in the area **206** to a different, second area **406** of the target object **116** as the area **206** of the ribbon **104** moves over this second area **406** of the target object **116**. The second area **406** does not overlap the first area **404** in the illustrated example. The ribbon **104** may not transfer the ink or pigment having another color in the area **204** or **208** to the second area **406** of the target object **116** as neither the area **204** nor the area **208** of the ribbon **104** move over the second area **406** of the target object **116**.

The ribbon **104** may be able to transfer the color of the ink in the area **208** to a different, third area **408** of the target object **116** as the area **208** of the ribbon **104** moves over this third area **408** of the target object **116**. The third area **408** does not overlap the first area **404** or the second area **406** in the illustrated example. The ribbon **104** may not transfer the ink or pigment having another color in the area **202** or **204** to the third area **408** of the target object **116** as neither the area **202** nor the area **204** of the ribbon **104** move over the third area **408** of the target object **116**.

Conversely, the ribbon **304** may transfer any of the inks or pigments in any of the areas **304**, **306**, **308** that extend across the width of the ribbon **304** to any area of the target object **116** (that the ribbon **304** moves over during the die strike). For example, the ink or pigment in the area **304** can be transferred to any of the areas **404**, **406**, **408**, the ink or pigment in the area **306** can be transferred to any of the areas **404**, **406**, **408**, and the ink or pigment in the area **308** can be transferred to any of the areas **404**, **406**, **408** because each of the areas **304**, **306**, **308** extends across the entire width of the ribbon **304**.

The ribbons **104**, **304** may be used to change the shading or color of the ink that is transferred from any of the areas **204**, **206**, **208**, **304**, **306**, **308**. For example, the same die **110** may repeatedly strike the ribbon **104**, **304** in one or more of the areas **204**, **206**, **208**, **304**, **306**, **308** to overprint the same location on the target object **116**. This overprinting can increase the amount of ink that is transferred to the target object **116**. The ink that is repeatedly transferred from the ribbon **104**, **304** may be from the same area **204**, **206**, **208**, **304**, **306**, **308** to change the shade (e.g., darken) of the color transferred to the target object **116**. With respect to the ribbon **304**, the die **110** may repeatedly strike different areas **304**, **306**, **308** of the ribbon **304** in the same location of the target object transfer and mix multiple colors in the same location or area of the target object **116**. For example, the die **110** can strike the area **304** in one location of the target object **116** and then strike the area **306** or **308** in the same location of the target object **116** to mix the color of the ink in the area **304** with the ink in the area **306** or **308** on the target object **116**.

The transfer of different colored inks or pigments to different locations on the target object **116** and/or overprinting onto the target object **116** can transfer multiple, different colors from the ribbon **104**, **304** to the target object **116** during a single pass of the ribbon **104**, **304** over the target object **116** and/or during a single strike of the die **110** on the target object **116**. For example, the ribbon **104**, **304** may be unwound from the unwind reel **102** to the wind-up reel **106** a single time while transferring multiple, different colors to the target object **116** (or multiple target object **116**).

The different colored inks or pigments can be transferred from the ribbon **104**, **304** using different dies **110**. For example, the die **110** can be changed for printing on different target objects **116** with the different dies **110** used to hot stamp print different information (e.g., different alphanumeric strings) using multiple, different colors.

FIG. **5** illustrates a flowchart of one embodiment of a method **500** for multi-color hot stamp printing. The method **500** can include operations described herein to transfer multiple, different colors in a hot stamp system using the ribbon **104**, **304** (or another ribbon having different areas with different colored inks or pigments). At **502**, a multi-color ribbon is obtained. As described above, this ribbon can have elongated areas having different colored inks or pigments, with the different areas extending end-to-end along the length of the ribbon. Optionally, the ribbon can have panels with different colored inks or pigments, with the different areas extending edge-to-edge along the width of the ribbon.

At **504**, the ribbon is moved over one or more target objects during a single pass. For example, the ribbon may be unwound from one reel and wound onto another reel a single time while passing over one or more target objects. At **506**, two or more of the different areas or panels of the ribbon are stricken by a die. This action transfers two or more different colors from the ribbon onto the target object. As a result, multiple, different colors can be printed onto each of one or more target objects during a single pass of the ribbon.

In one embodiment, a method is provided that includes moving a multi-color ribbon over a target object. The multi-color ribbon has different color inks or pigments in different areas of the multi-color ribbon. The method also includes applying heat and pressure onto the multi-color ribbon to transfer two or more of the different color inks or pigments to the target object during a single pass of the multi-color ribbon across the target object.

Optionally, the multi-color ribbon is moved across the target object in a web direction with the different areas of the multi-color ribbon being elongated along the web direction from one end of the multi-color ribbon to an opposite end of the multi-color ribbon along a length of the multi-color ribbon.

Optionally, the multi-color ribbon is moved across the target object in a web direction with the different areas of the multi-color ribbon extending from one edge of the multi-color ribbon to an opposite lateral edge of the multi-color ribbon.

Optionally, the heat and pressure are applied to transfer the two or more of the different color inks or pigments onto different locations of the target object.

Optionally, the heat and pressure are applied to change a shade of at least one of the different color inks or pigments that is transferred onto the same location of the target object multiple times during the single pass of the multi-color ribbon across the target object.

Optionally, the heat and pressure are applied to mix the two or more of the different color inks or pigments on the same location of the target object multiple times during the single pass of the multi-color ribbon across the target object.

In one embodiment, a multi-color ribbon configured for use in a hot stamp printing system is provided. The multi-color ribbon includes an elongated carrier that continuously extends from a first end to an opposite end and multiple, differently colored inks or pigments disposed in different areas of the elongated carrier between the first end and the second end of the elongated carrier. The elongated carrier is configured to be moved across a target object with heat and pressure applied to the elongated carrier to transfer two or more of the differently colored inks or pigments to the target object during a single pass of the elongated carrier across the target object for hot stamp printing of the two or more of the differently colored inks to the target object.

Optionally, the areas of the elongated carrier having the differently colored inks or pigments are elongated lanes that continuously extend from the first end to the second end of the carrier.

Optionally, the ribbon includes opposite first and second edges that each extend from the first end to the second end of the ribbon. The areas of the elongated carrier having the differently colored inks or pigments can be panels that continuously extend from the first edge to the second edge of the carrier.

Optionally, the differently colored inks are disposed in the different areas of the elongated carrier such that the two or more of the differently color inks or pigments are transferred onto different locations of the target object.

Optionally, the differently colored inks or pigments are disposed in the different areas of the elongated carrier such that the two or more of the differently colored inks are transferred onto the same location of the target object multiple times during the single pass of the multi-color ribbon across the target object.

Optionally, the differently colored inks or pigments are disposed in the different areas of the elongated carrier such that the two or more of the differently colored inks or pigments are transferred onto the same location of the target object multiple times during the single pass of the multi-color ribbon across the target object.

In one embodiment, a multi-color hot stamp ribbon is provided that includes an elongated carrier configured to be unrolled from a first reel and rolled onto a second reel of a hot stamp printing system and two or more different color inks or pigments disposed in one or more of different lanes or different panels of the carrier. The different color inks or pigments are located on the carrier such that two or more of the different color inks or pigments are transferred onto one or more of a common location or different locations of a target object that is printed on by the hot stamp printing system.

Optionally, the two or more different color inks or pigments are in the different lanes of the carrier.

Optionally, the different lanes of the carrier are elongated in directions that extend along a direction in which the carrier is unrolled from the first reel and rolled onto the second reel.

Optionally, the two or more different color inks or pigments are in the different lanes of the carrier such that the two or more different color inks or pigments are transferred onto different locations of the target object.

Optionally, the two or more different color inks or pigments are in the different panels of the carrier.

Optionally, the different panels of the carrier oriented in directions that extend perpendicular to a direction in which the carrier is unrolled from the first reel and rolled onto the second reel.

The singular forms "a", "an", and "the" include plural references unless the context clearly dictates otherwise. "Optional" or "optionally" means that the subsequently described event or circumstance may or may not occur, and that the description may include instances where the event occurs and instances where it does not. Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it may be related. Accordingly, a value modified by a term or terms, such as "about," "substantially," and "approximately," may be not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges may be identified and include all the sub-ranges contained therein unless context or language indicates otherwise.

FIG. **6** illustrates a cross-sectional view of another example of a multi-color ribbon **104** along line **6-6** shown in FIG. **2**. The ribbon **104** can include multiple layers **600**, **602** of different colored ink or pigments in at least one of the areas or lanes **204**, **206**, **208**. An upper or exposed layer **600** can be the ink or pigment that is visible along the top or exposed surface of the ribbon **104**, such as the layer that is visible in FIG. **2** in the area **204**, **206**, or **208**. A lower or covered layer **602** can be a different colored ink or pigment that is beneath the upper or exposed layer **600** and between the upper layer **600** and the elongated carrier **214**. For example, the upper layer **600** may be formed from or include red, green, white, or blue ink or pigments and the lower layer **602** can be formed from or include black ink or pigments. The lower or covered layer **602** may not be visible through the upper or exposed layer **600** in one embodiment.

In one embodiment, only one of the areas **204**, **206**, **208** includes the different colored layers **600**, **602**. Alternatively, two or more of the areas **204**, **206**, **208** include the different colored layers **600**, **602**. In an embodiment where two or more of the areas **204**, **206**, **208** include the different colored layers **600**, **602**, the underlying or covered layer **602** can be the same color in the different areas **204**, **206**, and/or **208** or may be different colors in different areas **204**, **206**, **208**. For example, the underlying layer **602** can be black ink or pigments beneath the red ink or pigments in the area **204** and the underlying layer **602** can be red, yellow, or white ink or pigments beneath the black ink or pigments in the area **206**.

FIG. **7** illustrates another example of the hot stamp printing system **100** shown in FIG. **1**. During printing using the multi-layered ribbon **104** having the different colored layers **600**, **602** in the hot stamp printing system **100**, the upper layer **600** may face the target object **116** such that the upper layer **600** is closer to the target object **116** and the carrier **214** is closer to the heated die **110**. The heated die **110** strikes the ribbon **104** and transfers at least some of the inks or pigments from the ribbon **104** to the target object **116** to be printed upon, such as an IV bag. This stamping onto the target object **116** transfers the pigments or inks that are in the ribbon **104** onto the target object **116**. This stamping also can transfer multiple layers of different colored inks or pigments onto the target object **116**. For example, the ink or pigment in the upper layer **600** can be transferred onto a surface of the target object and the ink or pigment in the underlying layer **602** can be transferred onto the ink or pigment from the upper layer **600**. This can result in the ink or pigment transferred to the target object **116** from the upper layer **600** being covered and hidden from view by the ink or pigment transferred to the target object **116** from the underlying layer **602**. For example, if the upper layer **600** includes black ink or pigments and the underlying layer **602** includes white ink or pigments, then the stamping of the heated die **110** on the ribbon **104** can transfer the black ink or pigments onto the surface of the target object **116** with the white ink or pigments on top of the black ink or pigments.

Alternatively, the multiple layers **806**, **808** can be transferred to the target objects **116** using another printing technique, such as gravure, screen printing, or the like.

FIG. **8** illustrates one example of an ablation assembly **800**. The ablation assembly **800** includes an energy source **802** that directs energy **804** toward the target object **116** to remove one or more upper or exposed layers **806** (or portions of layers **806**) of ink or pigments on the target object. For example, the energy source **802** can be a laser light source that emits a laser light toward the target object **116** that removes one or more portions of the ink or pigments stamped onto the target object **116** in an upper or exposed layer **806** to reveal the underlying ink or pigments on the target object **116** in an underlying layer **808**. The upper or exposed layer **806** of ink or pigments can be transferred onto the target object **116** from the lower layer **602** in the ribbon **104** while the lower or covered layer **808** of ink or pigments can be transferred onto the target object **116** from the upper layer **600** in the ribbon **104**.

For example, the ribbon **104** can be used to transfer the ink or pigment in the upper layer **600** shown in FIG. **6** onto a surface of the target object **106** and transfer the ink or pigment in the lower layer **602** shown in FIG. **6** on top of the ink or pigment on the target object **116** that was transferred from the upper layer **600**. The energy emitted by the energy source **802** can ablate or otherwise remove portions of the ink or pigment from the lower layer **602** (the visible ink or pigment on the target object **116**) to reveal the underlying ink or pigment from the upper layer **600** (which was previously not visible).

The ablation assembly **800** can be used to form variable data on the target object **116** using the layers **600**, **602** of inks or pigments transferred to the target object **116**. The hot stamp printing system **100** can transfer invariable information or data onto the target objects **116** and the ablation assembly **800** can remove part of the transferred inks or pigments to form variable information or data on the target object. For example, the same indicia (e.g., images, text, numbers, etc.) may be transferred onto several of the target objects **116** using the ribbon **104** with the layers **600**, **602** of ink or pigments in the ribbon **104**. This common indicia that is printed on several target objects **116** may be invariable data as the indicia does not change for multiple target objects **116**. The ablation assembly **800** can then remove different portions of the ink or pigments transferred onto the target objects **116** from the lower layer **602**. The portions of the ink or pigments that are removed can be different for each of two or more of the target objects **116** to reveal different portions of the underlying ink or pigments on the target objects **116**. The different revealed portions of the underlying ink or pigments can be variable data as the different revealed portions for different target objects **116** form different indicia. For example, the different revealed portions can be unique identifiers, bar codes, serial numbers, or the like, of the target objects **116**.

FIG. **9** illustrates a flowchart of one example of a method **900** for hot stamp printing. The method **900** can represent operations performed by the system **100** and the assembly **800** (which may be included in the system **100** or separate from the system **100**). At **902**, a multi-color ribbon is moved over one or more target objects. The ribbon can include overlapping layers of different colored inks or pigments, as described above. While one embodiment of the ribbon includes two layers of pigments or inks, alternatively, the ribbon may have three or more layers of overlapping or overlaying pigments or inks.

At **904**, the ribbon is stricken by a die. This action transfers two or more layers of different colors of inks or pigments from the ribbon onto the target object. At **906**, an upper layer of these different colored inks or pigments is at least partially removed by ablating the upper layer of inks or pigments transferred from the ribbon at **904**. This can reveal an underlying layer of a differently colored ink or pigment. The ablating can be individualized for the target objects by forming variable data or variable indicia, as described above.

In one embodiment, a method includes moving a multi-color ribbon over a target object. The multi-color ribbon has a lower layer and an upper layer of inks or pigments. The method also includes applying heat and pressure onto the multi-color ribbon to transfer the overlying layers of the inks or pigments to the target object, and removing at least a portion of the lower layer of the ink or pigment that was transferred to the target object to reveal at least a portion of the upper layer of the ink or pigment that were transferred to the target object.

Optionally, the lower layer and the upper layer of the inks or pigments are different colored inks or pigments.

Optionally, the lower layer of the ink or pigment in the multi-color ribbon is transferred to the target object as an upper layer and the upper layer of the ink or pigment in the multi-color ribbon is transferred to the target object as a lower layer.

Optionally, the at least the portion of the lower layer of the color inks or pigments is removed to form variable data on the target object.

Optionally, the at least the portion of the lower layer of the color inks or pigments is removed using laser ablation.

In one embodiment, a multi-color ribbon configured for use in a hot stamp printing system is provided. The ribbon includes an elongated carrier that continuously extends from a first end to an opposite end, and multiple, differently colored inks or pigments disposed in different layers of the elongated carrier between the first end and the second end of the elongated carrier. The elongated carrier is configured to be moved across a target object with heat and pressure applied to the elongated carrier to transfer two or more of the differently layers of the differently colored inks or pigments to the target object for hot stamp printing of the two or more layers of the differently colored inks to the target object.

In one embodiment, a hot stamp printing system includes a heated die configured to strike a multi-colored ribbon having overlying layers of different color inks or pigments to transfer the layers of the different color inks or pigments to a target object, and an ablation assembly configured to generate an ablation energy toward the layers of the different color inks or pigments on the target object, remove at least a portion of an upper layer of the layers of the different color inks or pigments on the target object, and reveal an underlying layer of the layers of the different color inks or pigments on the target object.

Optionally, the heated die is configured to strike the multi-colored ribbon such that the lower layer of the ink or pigment in the multi-color ribbon is transferred to the target object as an upper layer and the upper layer of the ink or pigment in the multi-color ribbon is transferred to the target object as a lower layer.

Optionally, the ablation assembly is configured to remove the at least the portion of the lower layer of the inks or pigments to form variable data on the target object.

This written description uses examples to disclose the embodiments, including the best mode, and to enable a person of ordinary skill in the art to practice the embodiments, including making and using any devices or systems and performing any incorporated methods. The claims define the patentable scope of the disclosure, and include other examples that occur to those of ordinary skill in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

What is claimed is:

1. A multi-color, hot stamp printing method comprising:
moving a multi-color ribbon over a target object in a single pass over the target object, the ribbon having different non-overlapping areas with each of the areas containing an ink or pigment of a different color than the other ones of the areas, each of the areas extending end-to-end along a length of the ribbon; and
striking two or more of the different areas in the ribbon with a die while the ribbon is moved over the target object in the single pass,
wherein the die strikes the two or more of the different areas to simultaneously transfer differently colored inks or pigments in the two or more of the different areas to the target object at the same time.

2. The multi-color, hot stamp printing method of claim **1**, wherein at least one of the areas of the ribbon includes lower and upper layers of the inks or the pigments, and striking the at least one of the areas of the ribbon with the die transfers the lower and upper layers to the target object, and further comprising:
removing at least a portion of the lower layer of the inks or the pigments that were transferred to the target object to reveal at least a portion of the upper layer of the inks or the pigments that were transferred to the target object.

3. The multi-color, hot stamp printing method of claim **2**, wherein the ribbon moves over the target object in a web direction that extends end-to-end along the length of the ribbon, and striking the ribbon with the die that is heated includes moving the die downward in a transfer direction to the ribbon.

4. The multi-color, hot stamp printing method of claim **3**, wherein striking the ribbon with the die that is heated uses heat and pressure to transfer the inks or the pigments to the target object.

5. The multi-color, hot stamp printing method of claim **4**, wherein striking the ribbon with the die transfers the inks or the pigments to an intravenous bag as the target object.

6. The multi-color, hot stamp printing method of claim **1**, wherein the ribbon includes two of the areas and striking the ribbon with the die includes striking both of the areas with the die.

7. The multi-color, hot stamp printing method of claim **1**, wherein the ribbon includes three or more of the areas and striking the ribbon with the die includes striking at least two of the three or more areas with the die.

8. The multi-color, hot stamp printing method of claim **1**, wherein striking the ribbon with the die transfers the inks or the pigments in a first area of the areas in the ribbon to a second area of the target object and transfers the inks or the pigments in a different, third area of the areas in the ribbon to a fourth area of the target object, but cannot transfer the inks or the pigments in the first area of the ribbon to the fourth area of the target object and cannot transfer the inks or the pigments in the third area of the ribbon to the fourth area of the target object.

9. The multi-color, hot stamp printing method of claim **8**, wherein the second area and the fourth area of the target object do not overlap each other.

10. The multi-color, hot stamp printing method of claim **9**, wherein striking the ribbon with the die cannot transfer the inks or the pigments in the first area of the ribbon to the fourth area of the target object and cannot transfer the inks or the pigments in the third area of the ribbon to the fourth area of the target object because moving the ribbon over the target object only moves the first area of the ribbon over the second area of the target object, only moves the third area of the ribbon over the fourth area of the target object, does not move the first area of the ribbon over the fourth area of the target object, and does not move the third area of the ribbon over the second area of the target object.

11. The multi-color, hot stamp printing method of claim **1**, wherein striking the ribbon with the die includes overprinting a same location on the target object with the inks or the pigments in the two or more areas of the ribbon by repeatedly striking the two or more areas of the ribbon with the tie to contact the same location on the target object.

12. The multi-color, hot stamp printing method of claim **11**, wherein striking the ribbon with the die to overprint the same location on the target object changes a shading of the inks or the pigments that are transferred to the target object.

13. The multi-color, hot stamp printing method of claim **1**, wherein the die is a first die that prints first information onto the target object by striking the two or more areas of the ribbon, and further comprising:

changing the first die to a second die to strike the ribbon; and

striking the ribbon with the second die to print second information onto the target object with the second information being different from the first information.

14. The multi-color, hot stamp printing method of claim **1**, wherein the multi-color ribbon is moved in a web direction extending end-to-end along the length of the ribbon while the two or more of the different areas in the ribbon are struck by the die.

15. A hot stamp printing method comprising:

moving a multi-color ribbon over a target object, the ribbon having an elongated carrier that continuously extends from a first end to an opposite end, the ribbon having different non-overlapping areas that each extend from the first end of the ribbon to the opposite end of the ribbon, the areas in the ribbon having differently colored inks or pigments on the elongated carrier; and

striking two or more of the areas in the ribbon with a die from a hot stamp printing system during a single pass of the ribbon over the target object to simultaneously transfer two or more of the differently colored inks or pigments to the target object at the same time.

16. The multi-color, hot stamp printing method of claim **15**, wherein at least one of the areas of the ribbon includes lower and upper layers of the inks or the pigments, and striking the at least one of the areas of the ribbon with the due-die transfers the lower and upper layers to the target object, and further comprising:

removing at least a portion of the lower layer of the inks or the pigments that were transferred to the target object to reveal at least a portion of the upper layer of the inks or the pigments that were transferred to the target object.

17. The multi-color, hot stamp printing method of claim **15**, wherein the ribbon includes three or more of the areas and striking the ribbon with the die includes striking at least two of the three or more areas with the die.

18. The multi-color, hot stamp printing method of claim **15**,

wherein striking the ribbon with the die transfers the inks or the pigments in a first area of the areas in the ribbon to a second area of the target object and transfers the inks or the pigments in a different, third area of the areas in the ribbon to a fourth area of the target object that does not overlap the second area of the target object, but cannot transfer the inks or the pigments in the first area of the ribbon to the fourth area of the target object and cannot transfer the inks or the pigments in the third area of the ribbon to the fourth area of the target object.

19. The multi-color, hot stamp printing method of claim **18**,

wherein striking the ribbon with the die cannot transfer the inks or the pigments in the first area of the ribbon to the fourth area of the target object and cannot transfer the inks or the pigments in the third area of the ribbon to the fourth area of the target object because moving the ribbon over the target object only moves the first area of the ribbon over the second area of the target object, only moves the third area of the ribbon over the fourth area of the target object, does not move the first area of the ribbon over the fourth area of the target object, and does not move the third area of the ribbon over the second area of the target object.

20. The multi-color, hot stamp printing method of claim **15**,

wherein striking the ribbon with the die includes overprinting a same location on the target object with the inks or the pigments in the two or more areas of the ribbon by repeatedly striking the two or more areas of the ribbon with the tie to contact the same location on the target object and to change a shading of the inks or the pigments that are transferred to the target object.

21. The multi-color, hot stamp printing method of claim **15**, wherein the die is a first die that prints first information onto the target object by striking the two or more areas of the ribbon, and further comprising:

changing the first die to a second die to strike the ribbon; and

striking the ribbon with the second die to print second information onto the target object with the second information being different from the first information.

22. The hot stamp printing method of claim **15**, wherein the multi-color ribbon is moved over the target object in a web direction that extends from the first end to the opposite end of the elongated carrier while the two or more of the areas in the ribbon are struck by the die.

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